

Cybersecurity Management

University of Minho

Master's Program in Cybersecurity

2025/2026

Practical Lab Exercises #04

I Questions

You are Junior Consultants at a large international consulting firm providing services to a multinational healthcare provider in the field of Risk Analysis. The company is in the process of adopting a new cloud-based patient management system that will host sensitive personal and medical information across multiple jurisdictions.

The Senior Consultant, your lead, will deliver a 15-minute presentation in two days to the company's Executive Board at the Burj Al Arab Jumeirah, in Dubai.

The Executive Board is composed of the Chief Executive Officer (CEO), Chief Financial Officer (CFO), Chief Information Officer (CIO), Chief Medical Officer (CMO), Chief Nursing Officer (CNO), Chief Operations Officer (COO), and Chief Compliance Officer (CCO).

The billionaire CEO dropped out of MIT's dual PhD program in Biomedical Engineering and Financial Technology and made her fortune in cryptocurrencies, living life at full throttle. She turned global healthcare into a playground for bold innovation, daring ventures, and calculated risk. A magnetic dealmaker, she surrounds herself with the world's sharpest minds and most skilled hands, forging unstoppable teams. Her motto: "Risk, luxury, and bold vision—because health is the ultimate wealth."

The CFO earned his Chartered Accountant designation in the UK and a Master's in Finance from INSEAD. He spent the early part of his career in investment banking before transitioning to healthcare finance, managing budgets and financial planning for several large hospital networks. He has over 15 years of experience in financial leadership, specializing in healthcare investment, insurance systems, and operational cost management. He frequently states to others: "Control the numbers, command the outcome."

The CIO holds a PhD in Information Systems from the University of Minnesota, a Master's degree in Health Informatics from the University of Edinburgh, and a Bachelor's in Computer Science from the University of Tokyo. She started her career as a software engineer in digital health solutions and later led IT departments in regional hospitals. She has over 10 years of experience in electronic health records, telemedicine, and security, with a

focus on integrating technology to improve patient care and operational efficiency. There is a pronounced and vigorous competition between the CEO and CIO, but it remains highly productive.

The CMO received his MD from Harvard University and completed his residency in internal medicine at the Mayo Clinic. He has over 25 years of clinical experience in both teaching hospitals and private practice. He has published research on chronic disease management and served on several advisory boards for medical policy and healthcare accreditation in the U.S. and Europe. Whispers suggest that his groundbreaking work in cardiothoracic surgery will, sooner or later, earn him a Nobel Prize.

The CNO holds a Bachelor's degree in Nursing from the University of Cairo and a Master's in Healthcare Leadership from Johns Hopkins University. She has worked for 15 years in advanced clinical settings, rising from staff nurse to nurse manager before becoming a director-level leader. She has extensive experience in patient safety initiatives, nursing education, and workforce development in private clinics and hospitals. She is a force of nature and an unwavering champion for patients. Her exceptional attentiveness and the personal care her team dedicates to patients command immense authority, granting her unparalleled influence over all the patients the company serves.

The COO earned a Bachelor's in Industrial Engineering from Purdue University and an MBA in Operations Management from HEC Paris. A retired Colonel with 10 years of military service, he honed his expertise in logistics and hospital operations before advancing to executive leadership overseeing multi-department healthcare facilities. He is a genius in operational efficiency, process optimization, and healthcare project management. Known as "The Fixer," the Colonel is a decisive and disciplined leader—unstoppable in driving operations and achieving results.

The CCO holds a Master's degree in European Union Law and a Bachelor of Laws degree from the University of São Paulo. He has 12 years of experience in healthcare regulation, working with both private clinics and government health authorities in Sweden, Germany, the United Kingdom, Canada, and Japan. Recognized for his expertise in risk management and regulatory compliance, above the door to his office reads: "Championing compliance, safeguarding care worldwide," a constant reminder of the standards he upholds every day. There is a charged yet courteous tension between the CCO and the COO, marked by mutual respect and high stakes.

The Senior Consultant sent an email asking you to prepare a five-slide presentation that he may use when addressing the Executive Board. The goal is to make a risk assessment that could serve as input for a decision on whether to move forward with the adoption of the cloud-based management system. In his email, he indicated that the first slide should focus on assumptions, the second on critical assets and valuation, the third on a ranked risk register, the fourth on a risk matrix, and the fifth on a reflection.

Attached to the email were some notes he had jotted about the company. These notes follow.

1. Customer base

Majority of customers from the USA, Canada, United Kingdom, Germany, France, Japan, Malaysia, Australia, Singapore, United Arab Emirates, Saudi Arabia, Qatar, India, Dubai, Hong Kong, Brazil, and Mexico

High-net-worth individuals, corporate executives, medical tourists, diplomatic personnel, and elite government officers

Company's headquarters located in Zurich, with regional centers in New York, London, Dubai, Singapore, Tokyo, Cape Town, and São Paulo

2. Workforce

6,000 medical professionals

12,000 nurses and care staff

4,000 Operations & Logistics personnel

3,000 Compliance, Legal, Finance, and Administration staff

2,500 R&D, Digital Health, IT staff

300 Executive and Regional leaders

3. Assets

Corporate ERP system at headquarters

Local servers and databases for patient management, billing, and human resources

Patient Records Database (approx. 4 million records; contains PII, financial, and sensitive health data)

Imaging Repository (approx. 6 million large files; sensitive health data)

Web Portal for Patients (used by 100,000 patients monthly; unclear if two-factor authentication is enabled for all)

Data Backup System (stored with third-party provider; contract details not fully available)

Internal Staff Access Platform (approx. 3,000 employees with varying access rights; there is evidence of significant "role creep")

Network Infrastructure (mix of on-premises and cloud-managed devices; some legacy firewalls still in place)

Mobile Application (planned rollout in 6 months; security audit not yet conducted)

4. Asset Values

Finance Department: estimates a patient record is worth \$200 in potential liability

Compliance Team: warns HIPAA and DGPR violations could result in fines up to \$50M per incident

Marketing: reputational loss of 20% drop in new patient acquisition in case of breach (no financial quantification)

Operations: believes system downtime of more than 2 hours could disrupt critical treatments in hospitals

5. Threat Intelligence

Last penetration test (18 months ago):

- Found SQL injection vulnerability in patient portal (report does not confirm if fixed)
- Weak password policies noted for internal staff accounts

Cloud vendors (6 months ago):

- Denial-of-service attacks on healthcare providers increasing by 35% year-over-year
- Average downtime cost estimated at \$15,000 per minute (no confidence interval given)

Industry report (current year):

- 75% of healthcare breaches originate from insider misuse or mistakes
- Estimated probability of ransomware in healthcare: 0.35 per organization per year

6. Likelihood Estimates

IT Security Team: rates ransomware likelihood as High (0.6 annually)

Cloud Vendors: one suggests ransomware likelihood 0.2 annually, the second 0.25, and the third 0.1

Independent Consultant: estimates ransomware likelihood 0.05 annually, citing better-than-average controls

No clear data on likelihood of natural disasters affecting cloud provider (though two data centers are located in flood-prone areas)

7. Impact Estimates

Average healthcare breach cost (IBM Report): \$11M per incident

Downtime of 24h estimated by Operations team: \$7.5M operational loss

Legal Counsel warns that noncompliance fines could stack if violations occur in multiple jurisdictions simultaneously

No estimate available for reputational harm beyond qualitative remarks

8. Providers

Company considering two cloud-based patient management systems: Oracle Health EHR and Epic Systems

Three potential providers for those systems (differences in security, compliance, technical support, scalability, long-term relationship viability, and cultural fit — analysis still underway)

2 Deliverable

The deliverable for this assignment must be submitted as a PDF document formatted for A4 paper size. Each page should be numbered with Arabic numerals, right-aligned in the footer.

The first page of the deliverable must include the following information: name of the course, name of the academic institution, academic year, title of the practical lab exercises, and identification of the students. Student identification should follow the structure: student number, m-dash, full name of the student, m-dash, program the student is enrolled. The identification of each student must appear on a separate line. Students in a group should be listed in alphabetical order by full name of the student.

The following five pages should provide the content for the five slides requested by the Senior Consultant (one slide per page).

Failure to follow these directives will result in a grade of zero for the assignment.

Each student must submit the deliverable individually, using the submission channel provided in the course community on the University of Minho's e-learning platform, by the deadline specified on that platform.

Students who do not submit the deliverable will receive a grade of zero. Submission attempts by other means or after the deadline will be rejected outright.

Prof. Filipe de Sá-Soares
University of Minho, 2025